

**Performance and Equity of Contextualized Geolocation Data for Lapse Prediction in
Alcohol Use Disorder**

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Abstract

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Introduction

About 1 in 10 adults in the United States met diagnostic criteria for alcohol use disorder (AUD) in 2022 (SAMHSA Center for Behavioral Health Statistics and Quality, 2022). Of these individuals, many will experience AUD as a chronic, relapsing disorder marked by periods of recovery interspersed with returns back to harmful use (McKay & Hiller-Sturmhofel, 2011; Moos & Moos, 2006). Lapses, or single instances of goal-inconsistent use, necessarily precede a relapse period (witkiewitzRelapsePreventionAlcohol2004b?). This temporal precedence, combined with their clear definition and ease of observation, make them a suitable target for early intervention. Yet, even when someone anticipates an oncoming lapse, it may be difficult to pinpoint its specific driving forces. Moreover, the precipitants to a lapse will vary between and within people. These two factors can make maintaining recovery goals difficult and motivate the need for lifelong monitoring and support.

One way to provide ongoing assistance to individuals in recovery is through the development of a continuous risk monitoring and support system. An ideal version of this system consists of two core components. First, the system must be able to collect risk-relevant data with sufficient temporal precision. Next, the system should communicate factors driving (or mitigating) this risk to provide personalized recommendations (e.g., behavior modification, seeking out supports). Not only does this system need to be developed outright, it also needs to be designed to be both sustainable (i.e., can be used over an extended period of time) and scalable

(i.e., can be effectively used by a maximal amount of people). These goals can be accomplished by integrating personal sensing data and machine learning within such a system.

Personal sensing data are data derived via embedded sensors in technology such as smartphones, smartwatches, or wearables (**mohrPersonalSensingUnderstanding2017a?**), and can also include information collected from applications downloaded onto one of these devices like ecological momentary assessment (EMA) surveys. Other examples of these data are location information, text messages, and social media behavior. Because these devices are already ubiquitous within daily life, these data can viably be collected unobtrusively and continuously for clinical purposes. Many of these data do not require individuals to significantly change their behavior or routines in any way, making data collection both sustainable and ecologically valid. Machine learning models can then uncover relationships between antecedent behaviors present in these data and true lapse events. Importantly, the use of machine learning models enables scalability. Continuous and long-term lapse risk detection and accompanying recommendations cannot realistically be provided by clinicians in an already overburdened addiction treatment system (McLellan et al., 2003). Irrespective of burden, *real time* risk detection and recommendations cannot reasonably be done by clinicians on a day-by-day basis.

Previous work has established the viability of brief daily surveys (also called “ecological momentary assessments” or EMAs) in predicting future alcohol use (**wyantMachineLearningModels2023?**). EMAs typically consist of questions pertaining to lapse-relevant constructs such as self-efficacy, past use, stress, and mood. However, EMAs cannot feasibly capture all contributors to lapse. Reliance on self-report is likely only an effective

tool for individuals who have a good level of insight and therefore are better able to identify these lapse precipitants. Moreover, completing several surveys per day might not be realistic for individuals with multiple jobs or comorbid conditions (e.g., depression). Predictions may be less reliable for individuals who provide either inaccurate or fewer responses.

Crucially, recovery is a dynamic process while EMA, by its very nature, collects data at discrete sampling periods. This may lead to a loss of information, as it assumes that responses remain constant between these periods (e.g., if positive mood is reported at one survey, no shifts in mood are captured until the next survey). Factors that contribute to both maintenance of recovery and lapse events change from person-to-person and from moment-to-moment. A shift in affect may precede a lapse for one individual but not another. Time spent experiencing craving may precede a lapse, but only after a certain threshold is met (i.e., a certain amount of time has been spent craving). In order to best capture this fluidity, the ideal information garnered from personal sensing to be used within a continuous risk monitoring and support system should be able to provide a correspondingly appropriate level of granularity. Geolocation data are one such promising source.

Geolocation Data for Risk Monitoring

Geolocation data consist of latitude and longitude coordinates and can be sampled continuously at regular intervals using applications on smartphones, and therefore have greater temporal sensitivity and specificity compared to EMA. Furthermore, the collection of these data requires little to no input from the user beyond initial set-up, while EMA necessitates repeated daily

engagement. In fact, many smartphones and smartwatches automatically collect geolocation data by default. Incorporating these data within a continuous risk monitoring and support system could mean reduced patient burden (i.e., fewer surveys to fill out) and a potentially more equitable system (i.e., easier for people who cannot fill out multiple surveys per day to engage with and, therefore, benefit from). Taken together, there is high potential for geolocation data to be feasibly harnessed both to improve upon past work and to eventually be integrated within a larger monitoring and support system.

Beyond the potential benefits geolocation data offer relative to EMA, the importance of location, such as environmental cues or one's perceived riskiness of a setting, has been shown to play an important role in lapse (Janak & Chaudhri, 2010; Walton et al., 1995; Walton et al., 2003). This link with lapse risk has led to the creation of coping skills that target substance-associated contexts in several treatment strategies like mindfulness-based relapse prevention (LeCocq et al., 2020). These findings underscore not only the potential wealth of information relating to lapse risk that an individual's location can provide, but also demonstrate the utility of incorporating location information into treatment. Furthermore, geolocation data have been identified as being of particular use in both understanding the precipitants to harmful substance use and its effective treatment (Stahler et al., 2013).

Research at the intersection of geolocation and mental health has generated a number of promising features which are derived from basic movement patterns (hereafter called "raw" features). These raw features quantify individual activity patterns as a way to examine behavioral change. For example, geolocation data have been used to estimate loneliness and isolation

through measures of circadian rhythmicity, movement speed, location variance, and clusters of frequently visited locations (Doryab et al., 2019). Many of these same features, in addition to others such as location entropy (the variability of time spent across significant location clusters), amount of time spent at home, and time spent in transit, have also been used to quantify symptoms of depression (Saeb et al., 2015; **saebMobilePhoneSensor2015b?**) and negative symptoms of schizophrenia (Raugh et al., 2020). This research has also linked experiential diversity in movement patterns to positive mood (Heller et al., 2020). These data have not only been applied to the measurement of mood symptoms, but to also predict their emergence (for review, see Shin & Bae, 2023).

In order to implement these features in the context of a monitoring and support system, it is important that these features be intervenable. In other words, the system has to be able to provide a recommendation to the individual using it. From an algorithmic perspective, it is important to balance both prediction and explanatory goals. Certain features detailed in the psychological literature, such as location variance and entropy, have clearer face validity in terms of intervenability as compared to others, like movement speed. For example, changes in the variability of someone's movement (e.g., location variance) might be indicative of significant life disruptions (e.g., losing one's job). In this scenario, encouraging the re-establishment of a regular routine to maintain recovery goals might be a helpful intervention point.

A benefit of these raw features is that they require little to no additional information on the part of both researchers and participants in order to identify meaningful movement patterns. While this makes them easy to implement, it does mean that these features are stripped of

important context that exists in our everyday environments. For example, two individuals could exhibit relatively regular movement patterns (like both attending work) while having a vastly different experience en route to those regular locations. For one individual, the route to the office might consist of coffee shops, greenspace, and affluent neighborhoods, whereas for another it might include bars, liquor stores, and higher crime areas.

Context-based features rectify this, in part, by capturing information about the qualities of the surrounding environment. In past work, geolocation data have been used to examine psychosocial stress exposure among substance users by using location data to obtain composite scores of community socioeconomic status and crime (Kwan et al., 2019). In a study of adolescents (both substance-using and not), Mennis and Mason leveraged domain expertise to identify local risky (e.g., pawn shops, corner stores, bars) and protective (e.g., recreational centers, after school programs) locations in a major U.S. city (Mennis & Mason, 2011). Yet, one can imagine further problems which might arise from simply pulling geographic features without individual-specific context. In the Mennis and Mason study, adolescents in this study, counter to researcher's predictions, consistently identified a given recreation center as risky because of the increased violence in that location through a structured qualitative interview ["Ecological Interview"; Mennis & Mason (2011)]. Thus, broad-strokes characterization of locations only based on domain expertise can obscure crucial person-specific differences.

Perhaps a middle ground between these approaches which leverages the benefit of personalized information seen in EMA with the relative low-burdensomeness of geolocation data is to ask participants to provide researchers with insight into what a given location means to

them. Geolocation features could be combined and further enriched with brief, intermittent surveys probing specific information about frequently visited locations (referred to as “contextualized” features). For example, geolocation and EMA data have been used to examine the relationship between location and mood in polydrug users (termed “geographical momentary assessment”; (Epstein et al., 2014)). Geolocation data have also been leveraged to alert individuals when they are approaching a self-identified risky location, such as a previously-frequented bar, within recovery-based smartphone applications (Attwood et al., 2017; Carreiro et al., 2021; Gustafson et al., 2014). Other recovery-based apps have expanded upon this by utilizing geolocation data to create “geo-fences” around areas of past use, like former smoking locations, such that individuals receive real-time notifications as they move through the environment (e.g., a pop-up message on a smartphone which reads “*You are entering a high-risk zone*”; (Naughton et al., 2016)).

Despite promising results suggesting that geolocation data can be capitalized on to improve our understanding of mental health outcomes generally and substance use patterns specifically, research has not been done to understand the *predictive* value of geolocation data in understanding lapses in AUD. Leveraging both raw and contextualized geolocation features, some of the more nuanced facets captured within location can be uncovered, such as associations with others (or lack thereof, e.g., social isolation), associations with previous or anticipated drinking behaviors (e.g., whether or not alcohol is present), and associations with affect (i.e., negative versus positive emotions tied to a given location). Using a combination of these insights in building a model for use within a continuous risk monitoring and support system will result in

our ability to identify a wider variety of potential lapse precipitants and, theoretically, more accurately capture heterogeneous experiences of recovery.

The Current Study

This study used geolocation data and machine learning to predict next-day lapse in individuals with a diagnosis of AUD and a recovery goal of abstinence in order to address several key gaps in the literature. First, we pursued a novel line of research by using geolocation data to predict lapses, expanding on previous risk monitoring work from our lab using EMA (Wyant Machine Learning Models 2023?). Second, we leveraged geolocation features from across the literature that are both raw and contextualized in order to cast a wide net across potential precipitants to lapse. To understand the utility of geolocation data broadly, we evaluated its performance against baseline characteristics (i.e., demographics). We further investigated what specific features were most predictive within our model to address our explanatory goal of uncovering relevant, actionable lapse risk factors which could be used to provide personalized recommendations in future work. This study constitutes an initial evaluation of a model designed to predict lapse back to alcohol use using minimally burdensome data that has the potential to be integrated within a continuous risk monitoring and support system.

Methods

Participants

One hundred and forty six individuals in early recovery (1-8 weeks of abstinence) from AUD were recruited from the Madison area to take part in a three-month study on how mobile health

technology can provide recovery support between 2017 and 2019 (R01 AA024391). Recruitment approaches included social media platforms (e.g., Facebook), television and radio advertisements, and clinic referrals. Prospective participants completed a phone screen to assess match with eligibility criteria, which necessitated that they were:

- ≥ 18 years of age
- Able to read and write in English
- Assessed as meeting criteria for moderate AUD (≥ 4 self-reported DSM-5 symptoms)
- Abstinent from alcohol for 1-8 weeks
- Willing to use only one smartphone while on study (personal or study-provided)

Participants were excluded if they exhibited severe symptoms of paranoia or psychosis (a score ≤ 2.24 on the SCL-90 psychosis scale or a score ≤ 2.82 on the SCL-90 paranoia scale administered at screening). Participants completed a baseline measure of demographics at the screening visit, which was used for fairness assessments and included: sex, race, ethnicity, employment status, income, and marital status.

Procedure

Participants enrolled in a three-month study consisting of five in-person visits, daily surveys, and continuous passive monitoring of geolocation data. Following screening and enrollment visits in which participants consented to participate, learned how to manage location sharing (i.e., turn off location sharing when desired), and reported frequently visited locations, participants completed three follow-up visits one month apart. At each visit, participants were asked questions about

frequently visited (>2 times during the course of the previous month) locations (Table 6).

Participants were debriefed at the third and final follow-up visit. Participants were expected to provide continuous geolocation data while on study. Other personal sensing data streams (EMA, cellular communications, sleep quality, and audio check-ins) were collected as part of the parent grant's aims (R01 AA024391).

To enable collection of geolocation data, participants downloaded either the Moves app or the FollowMee app during the intake visit. Moves was bought-out and subsequently deprecated while the study was ongoing (July 2018) and data collection continued using FollowMee until the end of the study. Both apps continuously tracked location via GPS and WiFi positioning technology. The FollowMee app uses adaptive sampling (samples location less frequently when the subject appears to be still), while the Moves app used fixed sampling (samples location steadily regardless of subject movement).

After completion of the study, data were processed to filter out duplicated points, fast movement speeds ($>100\text{mph}$), sudden positional jumps, and periods of long duration suggesting sampling error issues (>24 hours with no movement or >2 hours with a positional jump of more than 0.31 miles or 500 meters). Data points were classified as “in transit” when spacing between individual positions suggested a movement speed of greater than 4mph per NIH health guidelines (U.S. Department of Health and Human Services, 2018). Participants were considered to be at a known contextual location if they were within 0.031 miles (50 meters) of a reported frequently visited location. Start and end dates were adjusted for subjects in the case of high missingness

(e.g., if a subject was only adherent with GPS protocols for their last month on study, only their last month on study was retained).

Data analytic strategy

Data preprocessing, modeling, and Bayesian analyses were done in R using the tidymodels ecosystem (Kuhn & Wickham, 2020). Models were trained using high-throughput computing resources provided by the University of Wisconsin Center for High Throughput Computing (Center for High Throughput Computing, 2006).

In the interest of transparency, all analysis scripts and results can be found publicly on our study website (https://jjcurtin.gitub.io/study_gps). We additionally include the Transparent Reporting of a Multivariate Prediction Model for Individual Prognosis or Diagnosis extension for Artificial Intelligence (TRIPOD+AI) checklist in the supplement (Collins et al., 2024).

Outcome variable: Lapses

Alcohol lapses were used as the outcome variable in this study and were used to provide labels for model training, for testing model performance, and for testing issues of algorithmic fairness.

Future lapse occurrence (here conceptualized as next-day lapse) was predicted in 24-hour windows which advanced hour-by-hour. *Lapse* and *no lapse* occurrences were identified from the daily survey question, “*Have you drank any alcohol that you have not yet reported?*”.

Participants who responded *yes* to this question were then asked to report the date and hour of the start and the end of the drinking episode. In this case, the prediction window was labeled *lapse*.

Prediction windows were labeled *no lapse* if no alcohol use was reported within that window.

Input variables: Feature engineering

Feature engineering is the process of creating variables (or “*features*”) from unprocessed data and was used to transform raw geolocation data. Separate feature categories were created for the six contextualized geolocation categories (presented in Table 6) and for six raw geolocation categories: variability in location, time spent outside of the home in the evening, time spent at home, entropy, normalized entropy, and weather.

Raw geolocation features were selected from the literature on account of their ability to be tied to realistic recommendations in a continuous monitoring and support system, with the exception of weather which was included in an effort to increase precision (see ?@tbl-raw for a detailed description of feature calculations and potential application). Entropy and normalized entropy were calculated based on frequently visited locations (described above) instead of location clustering (e.g., Saeb et al. (2015)). Weather was calculated using the Applied Climate Information System (ACIS), which is maintained by the NOAA Regional Climate Centers (RCCs) using Weather FetchR, a collection of functions designed for querying this database and aligning it with geolocation data (Punturieri, 2025).

All contextualized features were calculated both as raw (i.e., summed duration over past 24 and 168 hour periods) and change features (i.e., relative to all previous geolocation data) in order to capture individual variation. The calculation windows, as well as whether raw and/or change features were calculated, for raw features are displayed in ?@tbl-raw.

Finally, we also created demographic features based on intake surveys: two quantitative (age, in years; income, in USD) and four dummy-coded (sex assigned at birth, male versus female; race/ethnicity, non-Hispanic White versus non-White and/or Hispanic; marital status, married versus not married versus other; education, high school or less versus some college versus some degree). Of our dummy-coded variables, education and marital status were one-hot encoded.

This resulted in a total of 108 features.

Imputation of missing data and removal of zero-variance features were additional general processing steps that were completed during feature engineering.

Algorithm development & performance

We trained and assessed several configurations of an XGBoost machine learning algorithm. The choice of using an XGBoost algorithm was motivated by two main reasons: 1) the calculation of Shapley values, used to understand the relative contributions of features in predictions, is optimized for XGBoost; and 2) previous work in our lab has made use of XGBoost algorithms in model development (**wyantMachineLearningModels2023?**) and the ability to eventually consolidate features across models in future work is of high priority.

We created two model types which varied across two feature sets: demographics only (baseline model) and a model which included all geolocation features as well as demographics (full model). This model comparison was designed to assess the value of geolocation features over demographics.

Configurations of models varied across a relevant and appropriate range of model-specific hyperparameters (trees, tree_depth, and mtry). No early stopping point was set and learning rate (eta) was set to .03 for all model configurations.

Models were trained and assessed using participant-grouped, nested k -fold cross-validation. Grouped cross-validation ensures that all data from a given participant are retained as either held-in or held-out, thereby preventing the introduction of bias from a participant's data being used to predict their own data. Nested cross-validation uses two nested loops for dividing and holding out folds: an outer loop, where held-out folds serve as test sets for model evaluation; and inner loops, where held-out folds serve as validation sets for model selection (Jonathan et al., 2000). We used 2 repeat of 5-fold cross-validation on the inner loop for model selection and 6 repeats of 5-fold cross-validation for evaluation.

The primary performance metric for model selection and evaluation of the validation sets was area under the Receiver Operating Characteristic (auROC) curve (Kuhn & Johnson, 2018). auROC indexes the probability that the model will predict a higher score for a randomly selected positive case (lapse) relative to a randomly selected negative case (no lapse). The overall percentage of lapses across all observations (each day per participant on study) was 7.4%, motivating the selection of this metric as it is unaffected by class imbalance.

In addition to providing information on the calibration of our model, we further calculate skill scores based on mean log loss. A skill score differs from proper scoring rules (such as the Brier score, Brier (1950)) in that it compares the performance of a model to a reference (Wheatcroft, 2019). Here, we compare the lapse base rate from the training data in each fold to

our model. This provides us with the relative changes in log loss than can be achieved by using a more complex model relative to merely predicting the lapse base rate.

Shapley values were computed in log-odd units in order to evaluate global importance of each feature category. Shapley values measure the unique contribution of features in an algorithm's predictions and therefore identify the relative importance of different features (Lundberg & Lee, 2017). Global feature importance for each broad feature category was calculated by averaging the absolute values of Shapley values across all observations per feature category. Highly important features represent relevant, actionable potential antecedents to lapse (and therefore points of intervention) that will be relevant in the future development of a fully integrated continuous risk monitoring and support system. However, these are descriptive analyses because standard errors or other indices of uncertainty for important scores are not available for Shapley values. Of note, even low global features may be an important contributor to lapse, and therefore motivate intervention, for a specific person at a specific time (i.e., high local importance).

Bayesian hierarchical generalized linear models were used to estimate the posterior probability distributions of auROCs. Median posterior probability for auROC and Bayesian credible intervals (CIs) are reported as an evaluation of each model's overall performance. We used a threshold of auROC = .5 (chance performance) when examining CIs, such that a CI that does not contain .5 indicates performance above chance and therefore predictive signal in the data.

Finally, Bayesian model comparisons were used to evaluate the model contrasts defined above based on feature sets. The probability that model performance systematically difference was determined for every model comparison. Posterior probabilities for auROC differences and the 95% Bayesian CI are reported.

Algorithmic fairness

Classes for fairness analyses were defined on the basis of personal individual characteristics divided such that subgroups reflected coarse dichotomies of groups which experience relatively increased and decreased societal privilege. This resulted in five broad classes: white versus non-white, younger than 55 versus equal to or older than 55, above or below the 2018 Federal poverty line (<https://aspe.hhs.gov/topics/poverty-economic-mobility/poverty-guidelines/prior-hhs-poverty-guidelines-federal-register-references>), educational attainment status (college or more or some college or less), and sex at birth (male versus female). A Bayesian hierarchical generalized linear model was used to estimate the posterior probability distributions of auROCs and corresponding 95% Bayesian CIs across these eight subgroups. Finally, Bayesian group comparisons were used in order to identify the likelihood of differential performance of our model between subgroups within each class.

Results

Demographics

A total of 192 individuals were eligible to participate in the study, of which 191 consented to participate and 169 enrolled in the study. Fifteen participants were excluded prior to the first

monthly follow-up visit. One participant was excluded for not maintaining a recovery goal of abstinence during their time on study. Two participants were excluded due to low adherence to study procedures. A further five individuals were excluded due to poor geolocation data quality from software incompatibility, resulting in a final sample size of 146. A detailed breakdown of participant characteristics is presented in Table 1.

Table 1: Demographic Characteristics

Characteristic	N = 146 ¹
Age	
18-24	10 (6.8%)
25-34	40 (27%)
35-44	41 (28%)
45-54	35 (24%)
55-64	17 (12%)
65+	3 (2.1%)
Sex At Birth	
Female	72 (49%)
Male	74 (51%)
education	
2-Year degree	12 (8.2%)
Advanced degree	23 (16%)
College degree	58 (40%)
High school or GED	13 (8.9%)
Less than high school or GED degree	1 (0.7%)
Some college	39 (27%)
work	
Disabled	6 (4.1%)
Employed	94 (64%)
Full-time student	7 (4.8%)
Homemaker	1 (0.7%)
Other, not otherwise specified	9 (6.2%)
Retired	8 (5.5%)
Temporarily laid off, sick leave, or	3 (2.1%)
maternity leave	
Unemployed	18 (12%)
Race	
American Indian/Alaska Native	3 (2.1%)
Asian	2 (1.4%)
Black/African American	7 (4.8%)
Other/Multiracial	7 (4.8%)
White/Caucasian	127 (87%)
Intercept	146 (100%)

¹ n (%)

auROC and Model Comparisons

The median auROC over all 30 folds was 0.72 (95% CI ([0.69, 0.74])). Figure 1 displays a receiver operating characteristic curve representing aggregate predicted lapse probabilities across all folds superimposed over each fold's individual curve. An additional histogram of model performance distribution across folds can be seen in Figure 2.

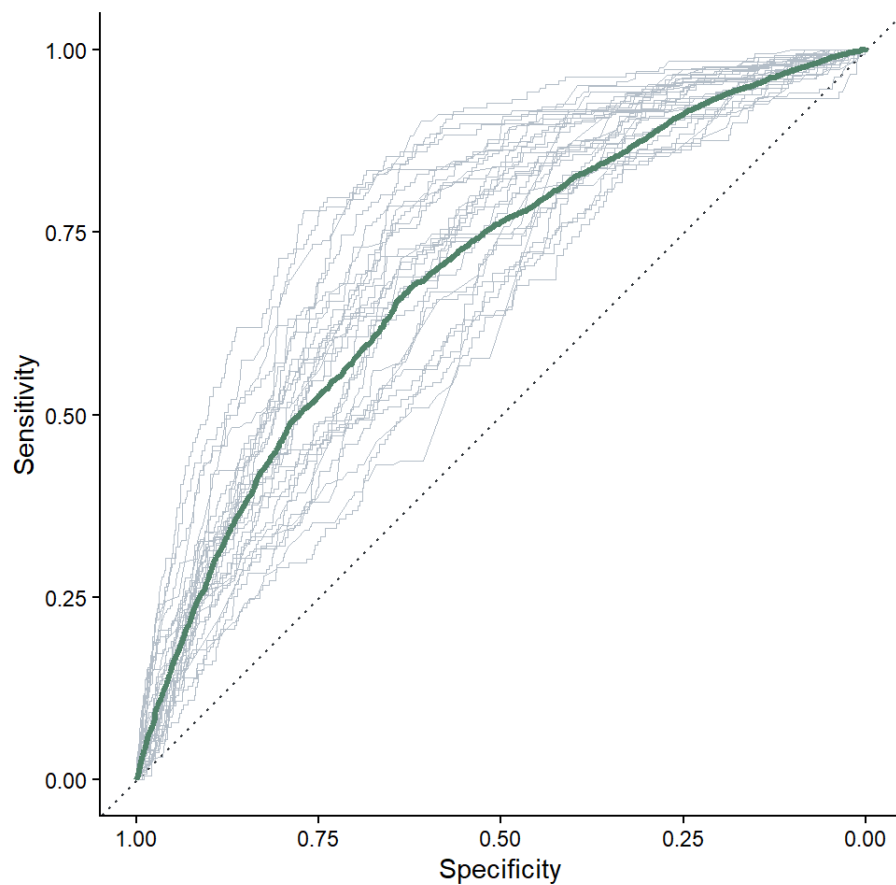


Figure 1: Area under the receiver operating characteristic curve (auROC) for the full model. An aggregated curve of model performance across 30 folds (dark green, bolded line) is superimposed over each fold's individual curve.

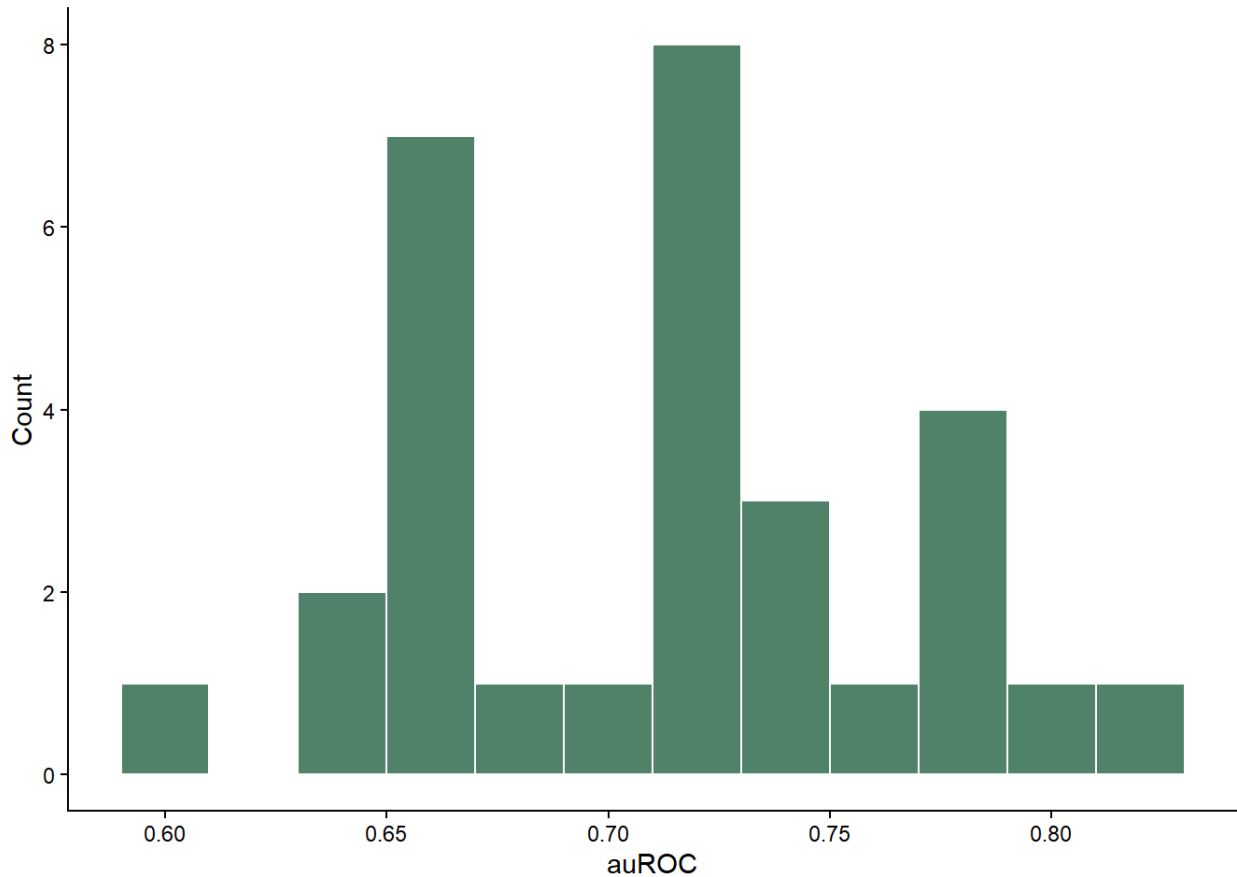


Figure 2: A histogram displaying median area under the receiver operating characteristic curve (auROC) for the full model across 30 folds.

To further characterize the predictive value of our geolocation features, we further compared the performance of our model to one which only included demographic features (0.52 (95% CI ([0.48, 0.55]))) as well as chance performance (.5). Compared to these more simplistic models, the model including geolocation features performed better (probabilities = 1).

Across fairness subgroup comparisons, we found evidence to suggest that median auROC differed by income and educational attainment, such that our model discriminates better (i.e., more accurately preserves relative ordering) in individuals who are younger than 55 compared to older (probability = (0.0019) and those with lower educational attainment compared to higher

educational attainment (probability = (0.9937) (Figure 3). A table of these comparisons can be seen in Table 2.

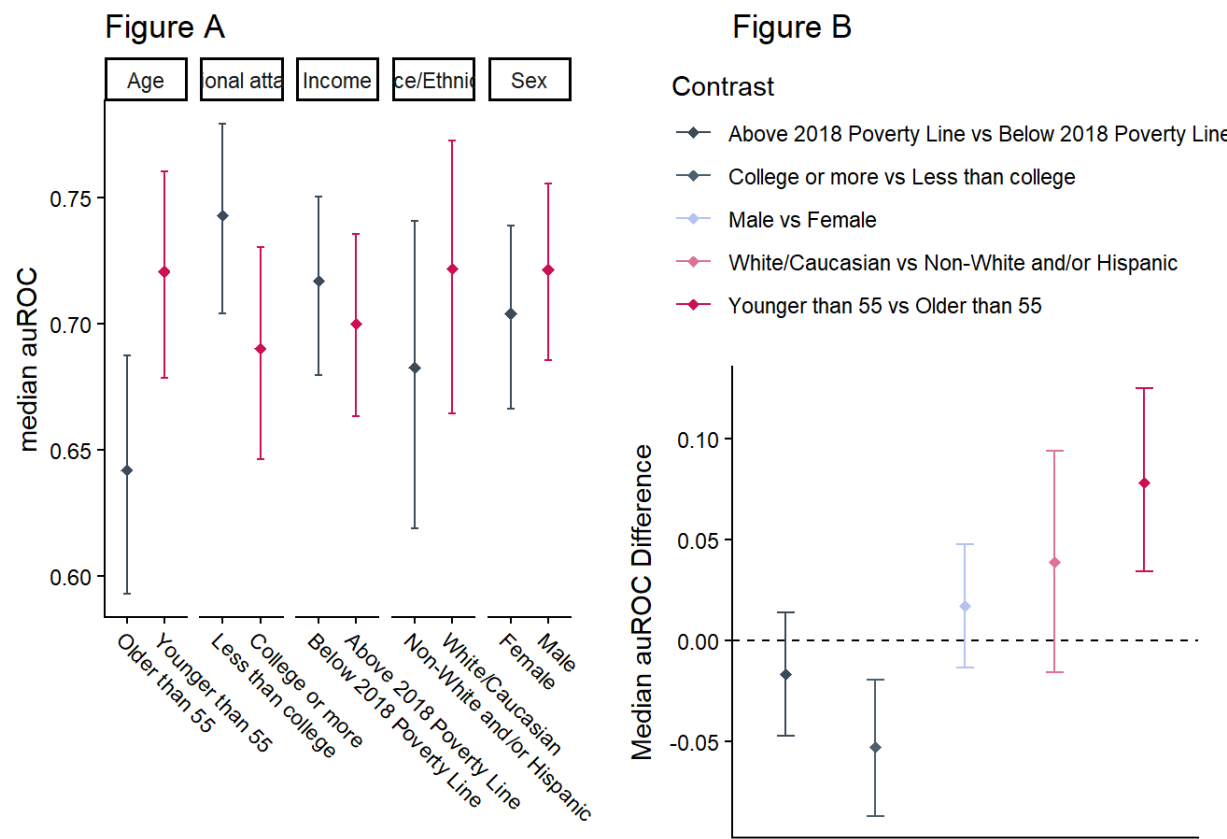


Figure 3: Bayesian comparisons of model performance by median auROC and median auROC difference.

Table 2: Bayesian comparisons of median auROCs by demographic subgroup

Contrast	Median auROC Difference	Bayesian CI	Probability
Male vs Female	0.017	[-0.013, 0.048]	0.169
White/Caucasian vs Non-White and/or Hispanic	0.039	[-0.015, 0.094]	0.119
Above 2018 Poverty Line vs Below 2018 Poverty Line	-0.016	[-0.047, 0.014]	0.810
Younger than 55 vs Older than 55	0.078	[0.034, 0.125]	0.002
College or more vs Less than college	-0.053	[-0.087, -0.019]	0.994

To further characterize model fairness, we additionally assessed whether median mean log loss differed across these same demographic groups. Our results suggest that our model produces less accurate probabilities (i.e., higher log loss) for individuals who are below the poverty line compared to those above (probability = (1.00) and women compared to men (probability = (1.00) (Figure 4). A table of these comparisons can be seen in Table 3.

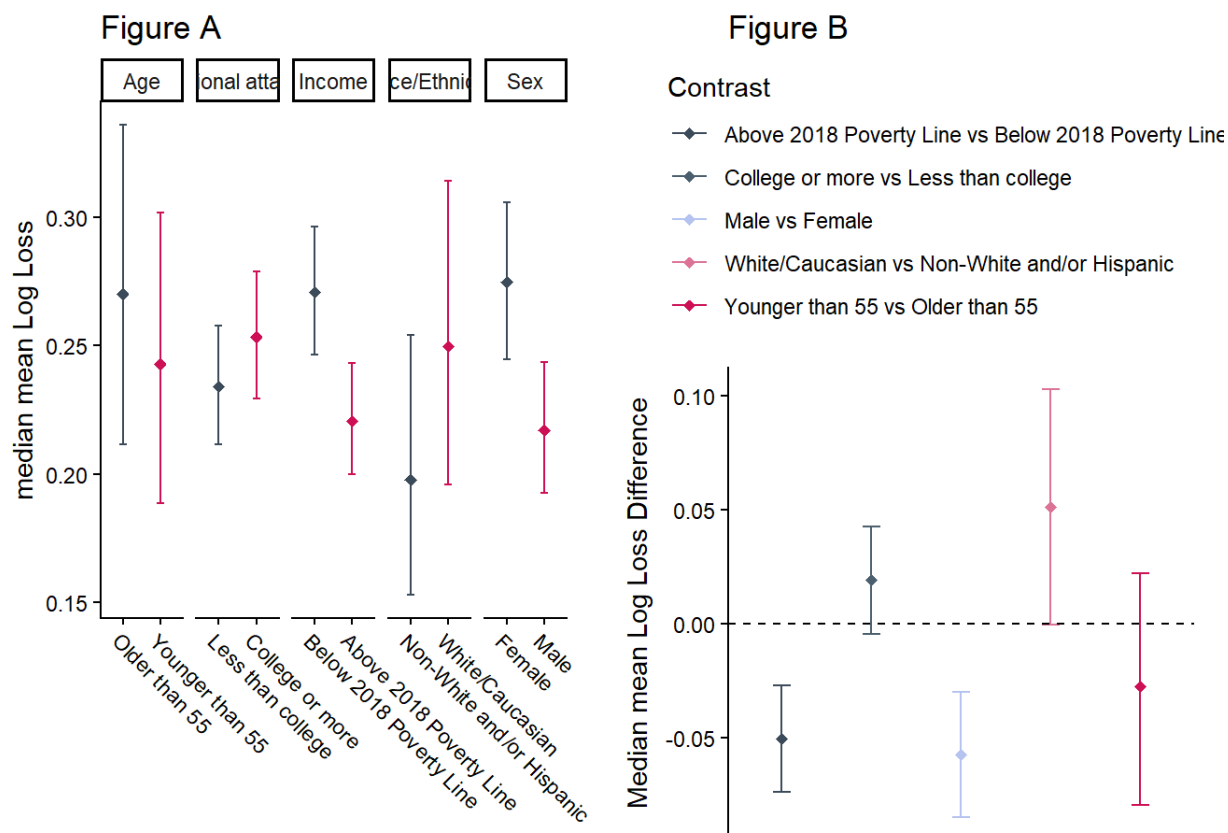


Figure 4: Bayesian comparisons of model performance by median mean Log Loss and median mean Log Loss difference.

Table 3: Bayesian comparisons of median mean Log Loss by demographic subgroup

Contrast	Median mean Log Loss Difference	Bayesian CI	Probability
Male vs Female	-0.057	[-0.085, -0.03]	1.000
White/Caucasian vs Non- White and/or Hispanic	0.052	[0, 0.103]	0.051
Above 2018 Poverty Line vs Below 2018 Poverty Line	-0.05	[-0.074, -0.027]	1.000
Younger than 55 vs Older than 55	-0.028	[-0.079, 0.022]	0.819
College or more vs Less than college	0.019	[-0.004, 0.043]	0.087

Calibration and Skill Scores

The best model configuration produced relatively well-calibrated probabilities which were clustered around the observed lapse rate. We present a calibration plot collapsed across all folds in Figure 5. The number of observations in each bin are presented in Table 4. We provide calibration plots and raw probability distributions separated by fold, along with a table displaying the number of observation in each bin, in Figure 6 and Table 5 in the supplement. For fidelity, bins with less than 50 observations are *not* included in the presented figures, but are available for reference in all tables.

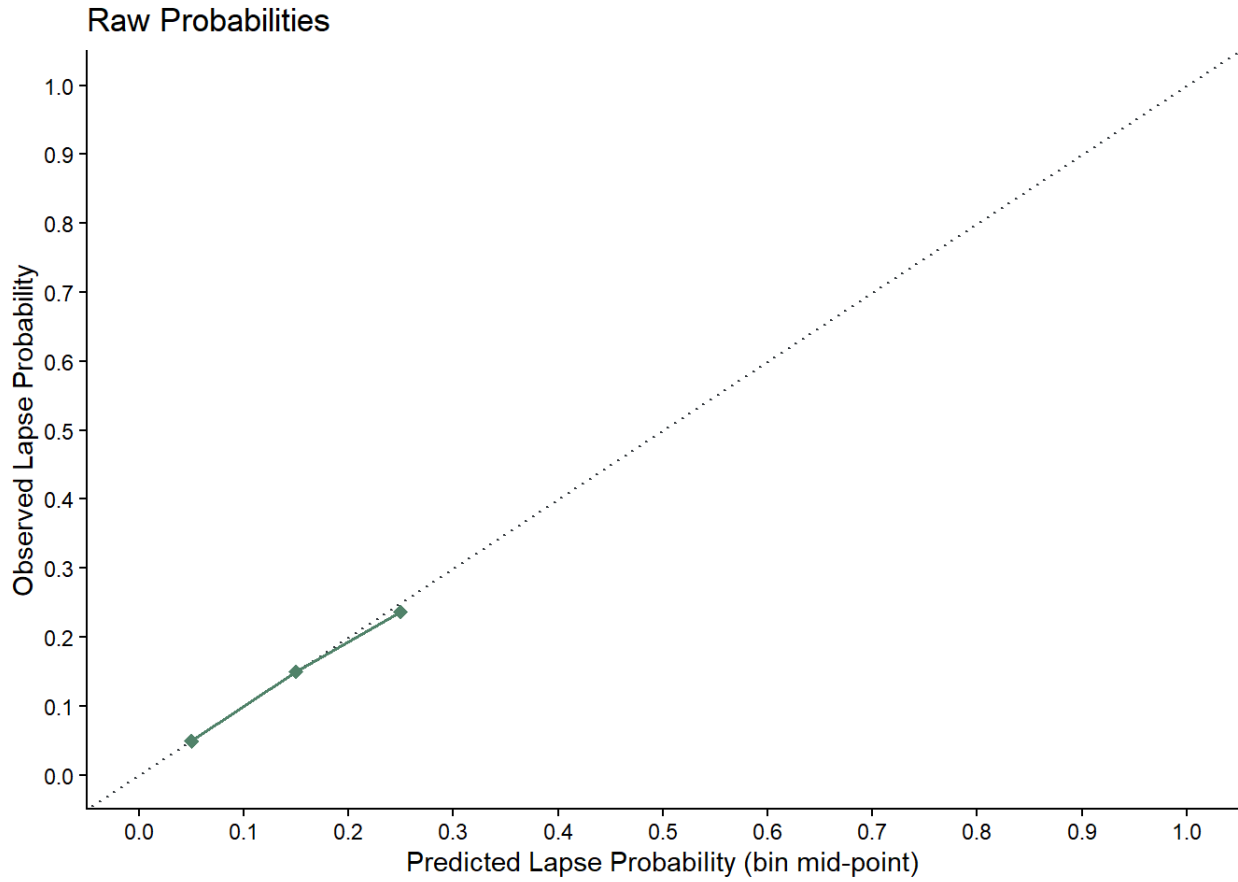


Figure 5: Calibration plot collapsed across folds (bins have $N \geq 50$ observations).

Table 4: Bayesian comparisons of median mean Log Loss by demographic subgroup

Bin width	Mean Probability of Lapse	Mean Lapse Percentage	Count
(0,0.1]	6.4	4.9	52192
(0.1,0.2]	12.8	14.9	15771
(0.2,0.3]	23.1	23.6	606
(0.3,0.4]	32.3	21.7	46
(0.4,0.5]	40.0	0.0	1

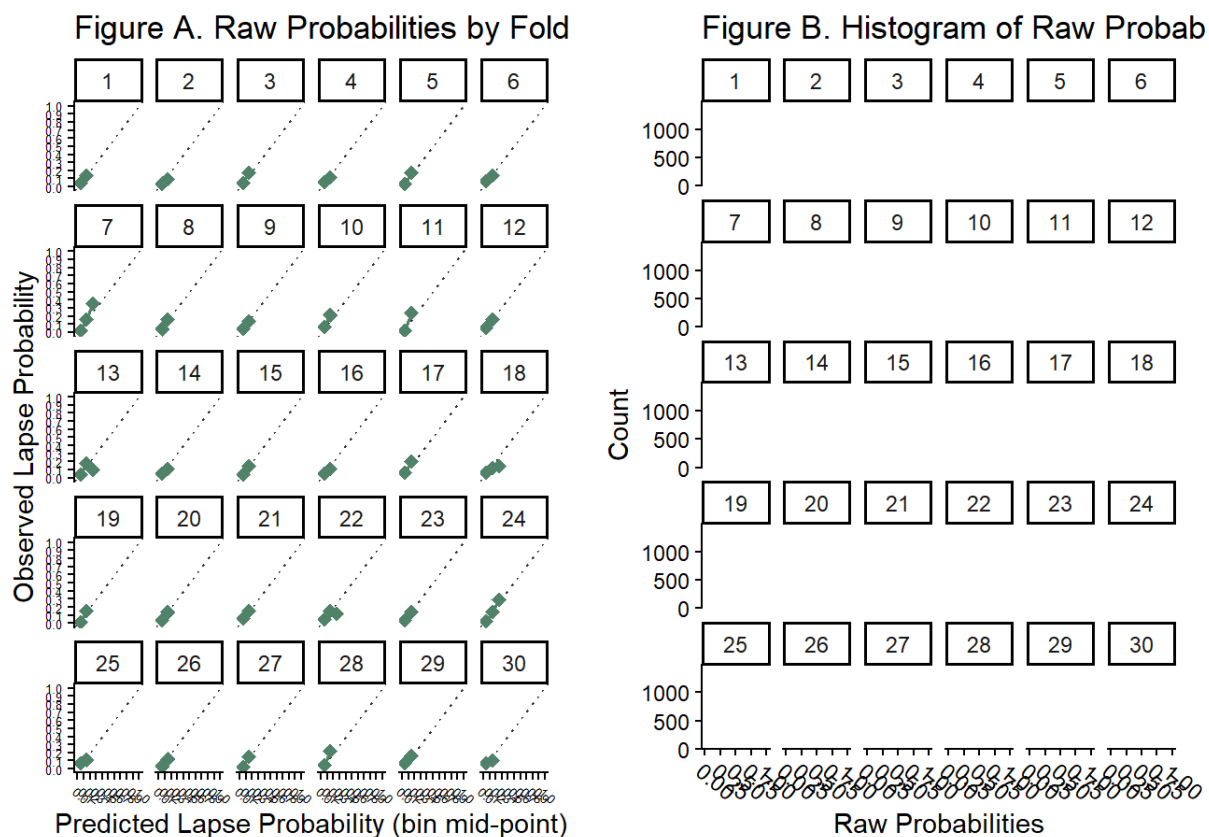


Figure 6: Raw probabilities displayed by fold, as shown by plotting predicted and observed lapse probabilities (A) and a histogram of predicted lapse probabilities (B).

Table 5: Bayesian comparisons of median mean Log Loss by demographic subgroup

Bin width	Fold	Mean Probability of Lapse	Mean Lapse Percentage	Count
(0,0.1]	1	6.5	5.1	1975
(0.1,0.2]	1	12.2	13.8	356
(0.2,0.3]	1	23.6	0.0	1
(0,0.1]	2	7.6	4.2	1433
(0.1,0.2]	2	13.7	9.3	841
(0.2,0.3]	2	23.1	33.3	3
(0,0.1]	3	7.7	4.5	1477
(0.1,0.2]	3	12.6	17.1	692
(0.2,0.3]	3	23.9	4.8	21
(0,0.1]	4	5.6	6.2	2117
(0.1,0.2]	4	12.2	11.3	291
(0,0.1]	5	8.0	3.5	1353
(0.1,0.2]	5	11.6	17.7	876
(0,0.1]	6	5.7	6.7	1781
(0.1,0.2]	6	12.9	13.8	600
(0.2,0.3]	6	21.7	37.0	27
(0,0.1]	7	7.3	1.9	1700
(0.1,0.2]	7	12.9	16.0	667
(0.2,0.3]	7	22.6	35.2	54
(0,0.1]	8	5.9	4.0	1786
(0.1,0.2]	8	12.8	15.9	389
(0.2,0.3]	8	21.4	0.0	6
(0,0.1]	9	5.5	4.7	1766
(0.1,0.2]	9	13.6	14.3	463
(0.2,0.3]	9	21.2	42.9	7
(0,0.1]	10	5.5	6.5	1899
(0.1,0.2]	10	13.7	21.3	258
(0.2,0.3]	10	21.6	24.2	33
(0,0.1]	11	8.0	2.7	1663
(0.1,0.2]	11	12.4	24.2	661
(0,0.1]	12	6.3	5.8	1793
(0.1,0.2]	12	12.0	15.9	496
(0.2,0.3]	12	20.5	0.0	1
(0,0.1]	13	5.8	3.9	1656
(0.1,0.2]	13	13.4	18.3	438
(0.2,0.3]	13	22.9	10.0	50
(0,0.1]	20	30.0	100.0	2000

The median skill score was (0.05, suggesting that mean log loss was reduced in our model relative to one which predicts the base rate of lapse.

Feature Importance

Global importance (mean absolute Shapley values) for the top 20 most important feature categories are displayed in [fig-7](#). Notably, the top three features were all related to self-reported risk (medium, high, and no). The most important demographic feature was income. There was an approximately three-fold difference in feature importance between the top performing geolocation feature as compared to the top performing demographic feature.

Discussion

Model Performance

Our day-level model of alcohol lapse prediction using geolocation data produced a median auROC of 0.72. Bayesian model comparisons indicated that this model performed better than both chance and a model which only included demographic variables. While not clinically implementable on its own, these findings confirm that there is strong evidence of predictive signal in these data and are an important first step towards an eventual continuous risk monitoring and support system. This study also provided explanatory insights by way of quantifying feature importance as well as a crucial examination of model fairness across classes, detailed in the following sections.

In general, our model was well-calibrated with predicted probabilities clustering around the observed event rate (i.e., lapse rate in our sample). As a result, additional calibration steps

were not necessarily despite our restricted range of probabilities. Importantly, this truncated range may not be an issue when anticipating future implementation. We are not necessarily interested in communicating exact probabilities to individuals about their lapse risk (such as, “*There is a 92% chance that you lapse back to use today*”). Rather, we are more interested in communicating *relative* levels of risk (such as, “*You are at a low risk level of lapse today,*” or, “*Your risk of lapse is higher this week compared to last week*” where *low* or *high* risk correspond to a designated probability threshold).

To better understand our model’s performance, we further examined model performance across coarsely divided subgroups (split by societal privilege) with respect to both median auROC and median mean log loss. We opted to leverage two approaches here as each results in a distinct on model output which reflect unique components of model fairness. For instance, we may be more concerned when relative ordering is reduced (as in our observation of reduced median auROCs in older adults or those with higher educational attainment) as compared to when probabilities are less accurate (as in our observation of higher mean log loss scores in women and those below the poverty line) if our goal is to ultimately communicate relative levels of lapse risk.

Of equal importance is to consider potential motivating factors behind these performance discrepancies. Reduced median auROCs may reflect features which do not equally capture the recovery experiences of individuals who are older adults or who have higher educational attainment. Phrased differently, our features may simply have more signal for younger adults and individuals with lower educational attainment. On the other hand, higher median mean log loss

may indicate that our model has a worse understanding of relative risk for women and those below the poverty line. Because we did not see differences in median auROCs for these groups, this suggests that our model can rank-order these individuals in terms of relative risk on par with their privileged counterparts. It is possible that these differences may also be due to differences in mobility patterns associated with sex, age, and income (which is frequently associated with one's educational attainment) Siła-Nowicka et al. (2016).

Feature Importance

We used Shapley values to quantify global feature importance. Overall, geolocation features were more important in model predictions than demographic variables, with time spent at medium, high, and no risk locations being the strongest contributors to risk predictions.

The most important demographic variable was income. This may be due to the fact that income can impact exposure to alcohol related cues in one's environment, with the potential for cues to drive increased craving and thus increase lapse risk. For instance, alcohol outlet density has been shown to be spatially related to income level (Morrison et al., 2016). These findings highlight the potential importance of personalizing feedback for a continuous risk monitoring and support system based, in part, on key demographic characteristics. For example, the approach for intervening on craving for an individual who cannot as readily avoid triggering environments may benefit from placing a greater emphasis on building self-confidence around resisting cravings as opposed to avoiding those spaces altogether, such as in the instance of an individual who lives in a neighborhood with a high liquor outlet density.

Though geolocation data provide a lower burden alternative to collecting important lapse-risk-relevant information as compared to EMA, we did survey our participants to provide us with their own contextual information about frequently visited locations. Some of our important features (such as time spent at restaurants or bars) could be automated and instead pulled from public map data to further reduce participant burden. Interestingly, our top four features (self-reported medium, high, or no risk and having previously drank alcohol at a location) may not be as easily garnered from public sources. For instance, what one person considers high risk to their recovery may not be high risk to another (e.g., what may be a town that one person passes through on their daily commute to another may be the location of an abusive ex-spouse). The same may be true for alcohol consumption. While public map data can assist in gathering information about liquor licenses, for instance, it cannot identify locations where individuals engaging in “secret drinking” (e.g., a flask in a car or at a place of work) might consume alcohol.

Overall, these results are well-aligned with the extant AUD literature, notably the focus of high-risk situations as an immediate determinant to relapse within the relapse prevention model (Larimer et al., 1999; Marlatt & Gordon, 1985).

Additional limitations and future directions

This study is an important step forward in building a model using geolocation data to predict next-day lapse back to alcohol use in individuals with a diagnosis of AUD and a recovery goal of abstinence. However, there are several limitations to this work.

First, our model has only a fair level of performance. A simple strategy to improve this is to explore additional features. Risk-relevant features that can be generated without additional participant-supplied contextual information include geographical spatial risk indicators and circadian rhythm data. Risk-terrain modeling is a spatial analysis technique which enables the estimation of environmental risk factors on a given outcome, such as contact with high liquor outlet density or high crime areas preceding lapse (Gajos et al., 2024). These features are not only clinically meaningful, but are also clinically *intervenable*. In the context of a continuous risk monitoring and support system, individuals could receive notifications relating to increasing amounts of time spent in riskier areas (beyond self-identified risky locations). Potential interventions in these examples would be to suggest relocating to a “safer” area.

Finally, a future direction of this work is to integrate the features engineering for this geolocation-based model with features from the EMA-based model designed within our lab (**wyantMachineLearningModels2023?**). These two studies represent foundational work carried out to explore predictive signal and important features across distinct data sources. The day-level EMA model performed better than the geolocation model, achieving an auROC of 0.90. However, both models appear to be contributing unique features of interest in predicting lapse which suggests that they may provide complementary information about lapse risk. For example, the top three day-level features using EMA were past use, future efficacy (i.e., belief in oneself to maintain recovery goals), and craving. Features explored across both of these models are being consolidated and used to optimize the development of a continuous risk monitoring and support

system for AUD where individuals will also receive temporally relevant feedback on features driving their lapse risk (Wyant et al., 2025).

Conclusion

This study demonstrated that geolocation data are able to predict next-day alcohol lapse at a fair level of performance ($\text{auROC} = 0.72$). Relative to our group's previously-developed EMA model (wyantMachineLearningModels2023?), geolocation data provide additional and unique contributing features to lapse risk predictions. While our model's level of performance is not sufficient for standalone implementation, these results suggest geolocation data are a viable supplement for a continuous risk monitoring and support system.

References

- Attwood, S., Parke, H., Larsen, J., & Morton, K. L. (2017). Using a mobile health application to reduce alcohol consumption: A mixed-methods evaluation of the drinkaware track & calculate units application. *BMC Public Health*, 17(1), 394. <https://doi.org/10.1186/s12889-017-4358-9>
- Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1), 1–3.
- Carreiro, S., Taylor, M., Shrestha, S., Reinhardt, M., Gilbertson, N., & Indic, P. (2021). Realize, Analyze, Engage (RAE): A Digital Tool to Support Recovery from Substance Use Disorder. *Journal of Psychiatry and Brain Science*, 6, e210002. <https://doi.org/10.20900/jpbs.20210002>
- Center for High Throughput Computing. (2006). *Center for high throughput computing*. Center for High Throughput Computing. <https://doi.org/10.21231/GNT1-HW21>
- Chen, M. K., & Pope, D. G. (2020). *Geographic Mobility in America: Evidence from Cell Phone Data* (Working {{Paper}} 27072). National Bureau of Economic Research. <https://doi.org/10.3386/w27072>
- Collins, G. S., Moons, K. G. M., Dhiman, P., Riley, R. D., Beam, A. L., Van Calster, B., Ghassemi, M., Liu, X., Reitsma, J. B., Van Smeden, M., Boulesteix, A.-L., Camaradou, J. C., Celi, L. A., Denaxas, S., Denniston, A. K., Glocker, B., Golub, R. M., Harvey, H., Heinze, G., ... Logullo, P. (2024). TRIPOD+AI statement: Updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*, 385, e078378. <https://doi.org/10.1136/bmj-2023-078378>
- Doryab, A., Villalba, D. K., Chikersal, P., Dutcher, J. M., Tumminia, M., Liu, X., Cohen, S., Creswell, K., Mankoff, J., Creswell, J. D., & Dey, A. K. (2019). Identifying Behavioral

Phenotypes of Loneliness and Social Isolation with Passive Sensing: Statistical Analysis, Data Mining and Machine Learning of Smartphone and Fitbit Data. *JMIR mHealth and uHealth*, 7(7), e13209. <https://doi.org/10.2196/13209>

Epstein, D. H., Tyburski, M., Craig, I. M., Phillips, K. A., Jobes, M. L., Vahabzadeh, M., Mezghanni, M., Lin, J.-L., Furr-Holden, C. D. M., & Preston, K. L. (2014). Real-time tracking of neighborhood surroundings and mood in urban drug misusers: Application of a new method to study behavior in its geographical context. *Drug and Alcohol Dependence*, 134, 22–29. <https://doi.org/10.1016/j.drugalcdep.2013.09.007>

Gajos, J. M., Gimenez-Santana, A., Walker, J. T., Cropsey, K. L., Walters, S. T., & Businelle, M. S. (2024). Using risk terrain modeling and geographically-explicit ecological momentary assessments to examine alcohol use in adults experiencing homelessness. *Journal of Crime and Justice*, 0(0), 1–14. <https://doi.org/10.1080/0735648X.2024.2396406>

Gustafson, D. H., McTavish, F. M., Chih, M.-Y., Atwood, A. K., Johnson, R. A., Boyle, M. G., Levy, M. S., Driscoll, H., Chisholm, S. M., Dillenburg, L., Isham, A., & Shah, D. (2014). A Smartphone Application to Support Recovery From Alcoholism: A Randomized Clinical Trial. *JAMA Psychiatry*, 71(5), 566. <https://doi.org/10.1001/jamapsychiatry.2013.4642>

Heller, A. S., Shi, T. C., Ezie, C. E. C., Reneau, T. R., Baez, L. M., Gibbons, C. J., & Hartley, C. A. (2020). Association between real-world experiential diversity and positive affect relates to hippocampal-striatal functional connectivity. *Nature Neuroscience*, 23(7), 800–804. <https://doi.org/10.1038/s41593-020-0636-4>

- Janak, P. H., & Chaudhri, N. (2010). The Potent Effect of Environmental Context on Relapse to Alcohol-Seeking After Extinction. *The Open Addiction Journal*, 3, 76–87. <https://doi.org/10.2174/1874941001003010076>
- Jonathan, P., Krzanowski, W. J., & McCarthy, W. V. (2000). On the use of cross-validation to assess performance in multivariate prediction. *Statistics and Computing*, 10(3), 209–229. <https://doi.org/10.1023/A:1008987426876>
- Kuhn, M., & Johnson, K. (2018). *Applied Predictive Modeling* (1st ed. 2013, Corr. 2nd printing 2018 edition). Springer. <https://doi.org/10.1007/978-1-4614-6849-3>
- Kuhn, M., & Wickham, H. (2020). *Tidymodels: A collection of packages for modeling and machine learning using tidyverse principles*.
- Kwan, M.-P., Wang, J., Tyburski, M., Epstein, D. H., Kowalczyk, W. J., & Preston, K. L. (2019). Uncertainties in the geographic context of health behaviors: A study of substance users' exposure to psychosocial stress using GPS data. *International Journal of Geographical Information Science*, 33(6), 1176–1195. <https://doi.org/10.1080/13658816.2018.1503276>
- Larimer, M. E., Palmer, R. S., & Marlatt, G. A. (1999). Relapse Prevention: An Overview of Marlatt's Cognitive-Behavioral Model. *Alcohol Research & Health*, 23(2), 151.
- LeCocq, M. R., Randall, P. A., Besheer, J., & Chaudhri, N. (2020). Considering Drug-Associated Contexts in Substance Use Disorders and Treatment Development. *Neurotherapeutics: The Journal of the American Society for Experimental NeuroTherapeutics*, 17(1), 43–54. <https://doi.org/10.1007/s13311-019-00824-2>
- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Proceedings of the 31st International Conference on Neural Information Processing Systems, NIPS'17*, 4768–4777.

Marlatt, G. A., & Gordon, J. R. (Eds.). (1985). *Relapse Prevention: Maintenance Strategies in the Treatment of Addictive Behaviors* (First edition). The Guilford Press.

McKay, J. R., & Hiller-Sturmhofel, S. (2011). Treating alcoholism as a chronic disease: Approaches to long-term continuing care. *Alcohol Research & Health: The Journal of the National Institute on Alcohol Abuse and Alcoholism*, 33(4), 356–370.

McLellan, A. T., Carise, D., & Kleber, H. D. (2003). Can the national addiction treatment infrastructure support the public's demand for quality care? *Journal of Substance Abuse Treatment*, 25(2), 117–121.

Mennis, J., & Mason, M. J. (2011). People, Places, and Adolescent Substance Use: Integrating Activity Space and Social Network Data for Analyzing Health Behavior. *Annals of the Association of American Geographers*, 101(2), 272–291. <https://doi.org/10.1080/00045608.2010.534712>

Moos, R. H., & Moos, B. S. (2006). Rates and predictors of relapse after natural and treated remission from alcohol use disorders. *Addiction (Abingdon, England)*, 101(2), 212–222. <https://doi.org/10.1111/j.1360-0443.2006.01310.x>

Morrison, C., Gruenewald, P. J., & Ponicki, W. R. (2016). Race, Ethnicity, and Exposure to Alcohol Outlets. *Journal of Studies on Alcohol and Drugs*, 77(1), 68–76. <https://doi.org/10.15288/jsad.2016.77.68>

Naughton, F., Hopewell, S., Lathia, N., Schalbroeck, R., Brown, C., Mascolo, C., McEwen, A., & Sutton, S. (2016). A Context-Sensing Mobile Phone App (Q Sense) for Smoking Cessation: A Mixed-Methods Study. *JMIR mHealth and uHealth*, 4(3), e106. <https://doi.org/10.2196/mhealth.5787>

Punturieri, C. E. (2025). *C-punturieri/weather-fetchr*.

- Raugh, I. M., James, S. H., Gonzalez, C. M., Chapman, H. C., Cohen, A. S., Kirkpatrick, B., & Strauss, G. P. (2020). Geolocation as a Digital Phenotyping Measure of Negative Symptoms and Functional Outcome. *Schizophrenia Bulletin*, 46(6), 1596–1607. <https://doi.org/10.1093/schbul/sbaa121>
- Saeb, S., Zhang, M., Kwasny, M., Karr, C. J., Kording, K., & Mohr, D. C. (2015). The relationship between clinical, momentary, and sensor-based assessment of depression. *2015 9th International Conference on Pervasive Computing Technologies for Healthcare (PervasiveHealth)*, 229–232. <https://doi.org/10.4108/icst.pervasivehealth.2015.259034>
- SAMHSA Center for Behavioral Health Statistics and Quality. (2022). *Highlights for the 2022 National Survey on Drug Use and Health*.
- Shin, J., & Bae, S. M. (2023). A Systematic Review of Location Data for Depression Prediction. *International Journal of Environmental Research and Public Health*, 20(11), 5984. <https://doi.org/10.3390/ijerph20115984>
- Siła-Nowicka, K., Vandrol, J., Oshan, T., Long, J. A., Demšar, U., & Fotheringham, A. S. (2016). Analysis of human mobility patterns from GPS trajectories and contextual information. *International Journal of Geographical Information Science*, 30(5), 881–906. <https://doi.org/10.1080/13658816.2015.1100731>
- Stahler, G. J., Mennis, J., & Baron, D. A. (2013). Geospatial technology and the "exposome": New perspectives on addiction. *American Journal of Public Health*, 103(8), 1354–1356. <https://doi.org/10.2105/AJPH.2013.301306>
- U.S. Department of Health and Human Services. (2018). *Physical Activity Guidelines for Americans, 2nd edition*.

Walton, M. A., Blow, F. C., Bingham, C. R., & Chermack, S. T. (2003). Individual and social/environmental predictors of alcohol and drug use 2 years following substance abuse treatment.

Addictive Behaviors, 28(4), 627–642. [https://doi.org/10.1016/s0306-4603\(01\)00284-2](https://doi.org/10.1016/s0306-4603(01)00284-2)

Walton, M. A., Reischl, T. M., & Ramanathan, C. S. (1995). Social settings and addiction relapse.

Journal of Substance Abuse, 7(2), 223–233. [https://doi.org/10.1016/0899-3289\(95\)90006-3](https://doi.org/10.1016/0899-3289(95)90006-3)

Wheatcroft, E. (2019). Interpreting the skill score form of forecast performance metrics.

International Journal of Forecasting, 35(2), 573–579. [https://doi.org/10.1016/j.ijforecast.2018.](https://doi.org/10.1016/j.ijforecast.2018.11.010)

11.010

Wyant, K., Sant’Ana, S. J., Punturieri, C. E., Yu, J., Fronk, G. E., Maggard, C. M., Janssen, C.,

Wanta, S. E., Kornfield, R., Swol, L. M. van, & Curtin, J. J. (2025). Maximizing Engagement,

Trust, and Clinical Benefit of AI-Generated Recovery Support Messages for Alcohol Use

Disorder: Protocol for an Optimization Study. *JMIR Research Protocols*, 14(1), e81697. [https://](https://doi.org/10.2196/81697)

doi.org/10.2196/81697

Tables and Figures

Table 1: Contextual geolocation information

Table 6: Location information collected from frequently visited locations.

Question	Responses
Address	
Type of place	Work, School, Volunteer, healthcare, Home of a friend, Home of a family member, Liquor store, Errands (e.g., grocery store, post office), Coffee shop or cafe, Restaurant, Park, Bar, Gym or fitness center, AA or recovery meeting, Religious location (e.g., church, mosque, temple), Other
Have you drank alcohol here before?	No, Yes
Is alcohol available here?	No, Yes
How would you describe your experiences here?	Pleasant, Unpleasant, Mixed, Neutral
Does being at this location put you at any risk to begin drinking?	No risk, Low risk, Medium risk, High risk
Did the participant identify this place as a risky location they are trying to avoid now that they are sober?	No, Yes

Table 2: Raw geolocation features

Feature	Original formula	Modification for current study	Interpretation	Theoretical mechanism of action
Location Variance	$\log(\sigma_{lat}^2 + \sigma_{long}^2)$	Calculated as both raw and difference scores for past 24 and 168 hours	Spatial variability over defined segments	Lower variance may be indicative of more isolation, less geographical coverage
Entropy	$Entropy = -\sum_{i=1}^N p_i \log(p_i)$	Calculated as both raw and difference scores for past 24 and 168 hours	Distribution of time across location clusters	Higher entropy may be indicative of greater flexibility in routine, which could either indicate experiential diversity or lack of routine
Normalized Entropy	$\frac{Entropy}{\log(N)}$	Calculated as both raw and difference scores for past 24 and 168 hours	Distribution of time across location clusters, accounting for number of clusters	
Time spent	Percentage of time spent at	Calculated as both raw and difference scores for past		

Whether, Percentage of CG, 24 and 168 hours

inspecting time spent at difference scores for past

: Raw geolocation features derived from the GPS literature with description of modifications
made for current study. {#tbl-raw}